

Assignment: ML (Regression)

Regression task:

- ➔ You will find Expanded_data_with_more_features dataset on the drive, use it to solve the assignment.
- ➔ Bonus solve this assignment using house_data dataset you will find it on the drive.

1.	Dataset Exploration:
	• Load the dataset and explore its structure. • Identify the target variable and potential features for linear regression.
2.	Data Preprocessing:
	• Handle missing values: Decide on an appropriate strategy (imputation, removal, etc.). • Explore and address outliers if present. • Check for and handle any categorical features using encoding techniques.
3.	Exploratory Data Analysis (EDA):
	• Visualize the distribution of the target variable. • Examine the relationships between features and the target through pair plots or correlation matrices.
4.	Feature Scaling:
	• Apply feature scaling if needed to ensure that all features are on a similar scale.
5.	Train-Test Split:
	• Split the dataset into training and testing sets.
6.	Linear Regression Model:
	• Build a linear regression model using a suitable library (e.g., scikit-learn). • Train the model on the training dataset.
7.	Model Evaluation:
	• Evaluate the model's performance on the testing set. • Use appropriate regression metrics (e.g., Mean Squared Error, R-squared).
8.	Visualization:
	• Visualize the predicted values against the actual values using scatter plots or regression plots.
9.	Conclusion:
	• Summarize the findings from the linear regression task. • Reflect on the model's performance and potential areas for improvement.